

Morphing Camber Wings

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Abstract—Birds achieve exceptional aerodynamic efficiency and maneuverability through smooth, continuous changes in wing shape, particularly camber and twist. In contrast, conventional aircraft rely on discrete control surfaces that introduce geometric discontinuities, increased drag, and mechanical complexity. Building upon a midterm investigation of curvature-driven camber morphing, this paper presents a completed study of a bio-inspired morphing wing using a reduced-order, two-dimensional elastic beam model representing a chord-wise airfoil cross-section.

The model incorporates dynamic morphing, spatially varying stiffness distributions, and simplified aerodynamic loading to investigate aero-elastic coupling during camber transitions. Five morphing configurations are simulated, including a baseline structural response, aerodynamic loading effects, biologically inspired curvature profiles, tip-softened stiffness, and linearly tapered stiffness representative of real aircraft wings. Simulation results quantify deformation magnitude, internal bending energy, and aerodynamic reaction forces. The findings demonstrate that curvature shaping and stiffness gradients strongly influence morphing effectiveness and energetic cost, while aerodynamic loading significantly increases internal energy without substantially altering final geometry. This framework provides a computationally efficient foundation for future three-dimensional twist-camber morphing and feedback-controlled aero-elastic simulations.

I. PROBLEM STATEMENT AND MOTIVATION

Conventional aircraft and unmanned aerial vehicle (UAV) wing designs rely heavily on discrete control surfaces such as flaps, ailerons, and elevators to manipulate airflow and generate lift or control moments. While effective, these mechanisms introduce geometric discontinuities that promote flow separation, increase drag, and add mechanical complexity. These drawbacks are particularly detrimental at the low Reynolds numbers typical of small UAVs, where even minor surface discontinuities can significantly degrade aerodynamic performance. In addition, actuator-based control systems increase system mass and introduce failure modes that limit robustness and scalability.

In contrast, birds and other flying animals modulate their aerodynamic surfaces through smooth, continuous changes in wing shape. They can dynamically adjust camber, span, and twist in response to external conditions without discrete control surfaces. For example, swifts and falcons actively control wing curvature and twist during glide and maneuvering to optimize lift-to-drag ratio, delay stall, and respond to gusts [1]. This biologically-inspired approach to morphing suggests a path forward for low-mass, high-agility UAV designs.

The objective of this work is to investigate continuous camber morphing using a reduced-order structural model

inspired by biological wings. The wing is represented as a two-dimensional elastic beam corresponding to a chord-wise airfoil cross-section capable of smooth shape change through curvature-based actuation. This modeling approach enables systematic study of how prescribed curvature distributions, stiffness gradients, and aerodynamic loading influence morphing behavior. By focusing on structural response and aero-elastic interaction rather than high-fidelity aerodynamics, the present study aims to identify key design parameters governing camber morphing effectiveness and energetic cost.

II. BACKGROUND AND LITERATURE REVIEW

Wing morphing has long been a topic of interest in both biological flight studies and aerospace engineering. Birds and bats achieve remarkable aerodynamic versatility through continuous geometric transformations of their wings, dynamically adjusting span, camber, and twist to optimize performance across a wide range of flight conditions. These smooth deformations allow biological flyers to enhance lift-to-drag ratio, delay stall, and respond effectively to gusts without relying on discrete control surfaces.

Biological studies have shown that birds actively control wing camber to improve aerodynamic efficiency. Lentink *et al.* demonstrated that swifts modulate camber during gliding to enhance performance despite structurally fixed wings [1]. These shape changes are achieved through a combination of musculoskeletal structure, feather articulation, and passive material compliance rather than discrete joints or actuators.

In aerospace engineering, traditional morphing concepts often rely on embedded actuators, hinges, and linkages [2]. While effective at larger scales, these approaches introduce additional weight, structural complexity, and potential failure points. To address these limitations, researchers have explored compliant mechanisms and smart materials such as shape memory alloys (SMAs) and piezo-electrics to enable distributed deformation with fewer moving parts [3], [4]. Reviews by Sofla *et al.* highlight the broader challenges associated with morphing wing design, including actuation complexity, material limitations, and aero-elastic coupling [5]. In addition, Thill *et al.* emphasized the importance of morphing skin technologies capable of sustaining aerodynamic loads while allowing smooth, continuous deformation [6].

Reduced-order structural models have been widely used to study morphing behavior in an efficient computational setting. Elastic beam and plate models are particularly effective for

representing camber deformation, while torsional rod and shell models can capture spanwise twist [7]. Such models enable rapid parametric studies and provide insight into structural behavior that may be obscured in high-fidelity simulations.

Morphing structures inherently involve coupling between structural deformation and aerodynamic loading. Aero-elastic effects can significantly influence both achievable morphing shapes and energetic cost. While many prior studies focus either on passive aero-elastic deformation or on high-fidelity coupled simulations, reduced-order approaches allow first-order aero-elastic interactions to be examined efficiently. Control of compliant morphing structures presents additional challenges due to nonlinear and underactuated dynamics, and prior work has explored feedback-based control strategies for regulating deformation [8], [9]. Experimental and numerical studies have also demonstrated the feasibility of SMA-actuated adaptive wings capable of producing smooth camber and twist variations under aerodynamic loading [10]. In contrast, the present study emphasizes curvature-based actuation and passive stiffness tailoring to examine dynamic morphing and aero-elastic effects without explicit feedback control.

III. METHODS

A. Structural Model Formulation

The morphing wing is modeled as a two-dimensional elastic beam representing a chord-wise cross-section of an airfoil. The beam is discretized into twenty-one nodes connected by elastic segments, allowing large bending deformations while maintaining numerical stability. Each segment is characterized by axial stiffness, bending stiffness, and mass properties consistent with a lightweight composite wing structure. Material parameters are selected to approximate carbon-fiber-like behavior, with a Young's modulus of 200 GPa and a density of 1700 kg/m³. A complete list of simulation parameters, including geometric, material, and aerodynamic properties, is provided in Appendix A. The beam root is fully fixed in position and orientation, while the first interior node is dynamically displaced according to the local prescribed curvature. The remainder of the beam is free to deform under internal actuation and external aerodynamic loading.

This reduced-order formulation enables efficient simulation of camber morphing while capturing the dominant structural mechanics governing chord-wise deformation. Stretching deformation is allowed but remains negligible in all cases, confirming that the system operates in a bending-dominated regime.

B. Curvature-Based Camber Actuation

Camber morphing is achieved through prescription of a spatially varying natural curvature distribution, $\kappa_0(s)$, along the beam length. Rather than applying explicit moments or forces at discrete locations, the beam is driven toward a target shape by minimizing the bending energy associated with deviations between the actual curvature $\kappa(s)$ and the

prescribed curvature field. The bending energy is expressed as

$$E_b = \frac{1}{2} \int EI(s) (\kappa(s) - \kappa_0(s))^2 ds, \quad (1)$$

where $EI(s)$ denotes the local bending stiffness.

Several curvature distributions are implemented to explore both engineering-inspired and biologically inspired camber shapes. These include uniform curvature, linear curvature, exponential curvature concentrated toward the trailing edge, and Gaussian curvature profiles. To ensure physically realistic deformation, the magnitude of the prescribed curvature is limited to a maximum value of $\kappa_0 = -0.1$. Dynamic morphing is simulated by smoothly ramping the curvature from zero to its target distribution over a 1-second transition period. This time-dependent ramping is applied as a linear scaling factor on the spatial curvature profile and ensures gradual actuation, avoiding unrealistic transient effects caused by abrupt changes in loading.

C. Variable Stiffness Modeling

To investigate the role of passive structural compliance, spatially varying bending stiffness distributions are incorporated into the beam model. In addition to a uniform stiffness baseline, two nonuniform stiffness profiles are examined. A tip-softened configuration reduces bending stiffness near the trailing edge, while a linearly tapered configuration decreases stiffness gradually from the root toward the tip, reflecting common aircraft wing structural layouts.

These stiffness variations are applied multiplicatively to the baseline stiffness distribution, allowing deformation sensitivity to vary along the beam without altering overall material properties. This approach enables assessment of how stiffness gradients influence achievable camber, internal energy storage, and aero-elastic response.

D. Aerodynamic Force Model

Aerodynamic effects are included using simplified distributed force models to capture first-order aero-elastic coupling. Two aerodynamic regimes are considered. In the baseline structural case, no aerodynamic forces are applied, allowing isolation of the pure structural response to curvature actuation. In aerodynamic cases, a high Reynolds number drag model is employed with a uniform freestream velocity of 20 m/s. Aerodynamic forces act opposite to the local velocity of each beam segment, generating distributed drag and normal forces that resist deformation.

Although the aerodynamic model is intentionally simplified, it provides sufficient fidelity to assess the energetic cost of morphing under airflow and to quantify the influence of aerodynamic resistance on deformation magnitude and reaction forces.

E. Numerical Integration and Simulation Procedure

The equations governing beam deformation are integrated forward in time using an implicit time-stepping scheme to ensure numerical stability under large deformations. At

each time step, internal elastic forces arising from bending and stretching are computed alongside external aerodynamic forces. The resulting system of equations is solved iteratively until convergence is achieved.

Each simulation begins from an initially straight, undeformed configuration. The prescribed curvature field is then ramped from zero to its target distribution, producing smooth dynamic camber morphing. Throughout the simulation, internal energies, nodal positions, and reaction forces at the fixed root are recorded. These quantities are used to compare morphing performance across curvature profiles, stiffness distributions, and aerodynamic loading conditions.

IV. SIMULATION CASES

The following dynamic morphing cases are simulated:

- **Case A:** Baseline morphing with uniform stiffness, exponential curvature profile, and no aerodynamic loading.
- **Case B:** Uniform stiffness with exponential curvature under high Reynolds number aerodynamic loading.
- **Case D:** Uniform stiffness with a Gaussian curvature profile under aerodynamic loading.
- **Case E:** Tip-softened stiffness with an exponential curvature profile under aerodynamic loading.
- **Case H:** Linearly tapered stiffness with a linear curvature profile under aerodynamic loading.

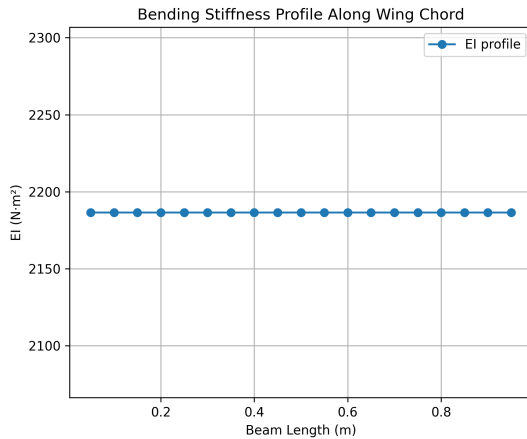


Fig. 1. Bending stiffness (EI) profiles for Cases A, B, and D — all uniform. Non-uniform profiles are shown later (e.g., in Case H).

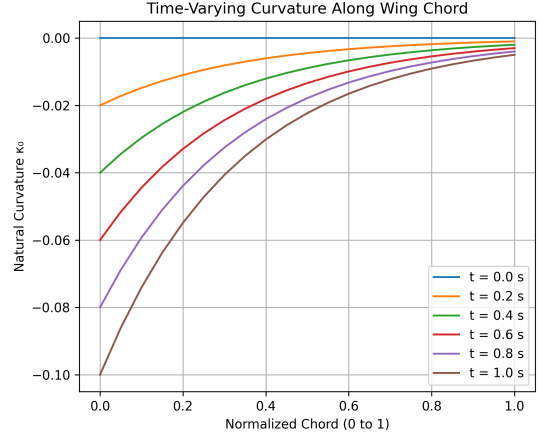


Fig. 2. Exponential κ_0 profile used in Cases A, B, and E. Distinct profiles for D and H appear in their respective sections.

V. RESULTS AND CASE COMPARISONS

A. Case A — Baseline Structural Response (No Drag)

This baseline case isolates the structural response of the morphing beam by excluding aerodynamic forces. A uniform bending stiffness profile is applied alongside an exponential curvature input ramped over 1 second. The beam morphs solely due to internal actuation, providing a reference for comparison against drag-influenced cases.

The beam exhibits smooth and symmetric deformation toward the target shape. With no aerodynamic resistance, the reaction forces arise purely from the curvature-induced motion. Internal energy is dominated by bending, peaking early in the transition before settling into a low steady-state value.

Observations:

- **Final tip displacement:** $\Delta x = -0.097$ m, $\Delta y = -0.395$ m (total: 0.406 m)
- **Final internal energy:** Bending = 5.9×10^{-5} J; Stretching = 0 J
- **Peak bending energy:** 2.64×10^{-3} J
- **Reaction forces at root:** Drag = -0.095 N; Lift = -22.03 N
- **Net lift margin:** -19.39 N

Figures 3–5 visualize the deformation, force response, and energy evolution.

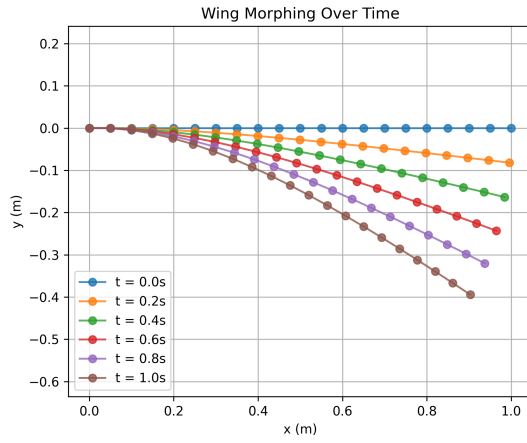


Fig. 3. Case A — Deformation over time due to exponential curvature.

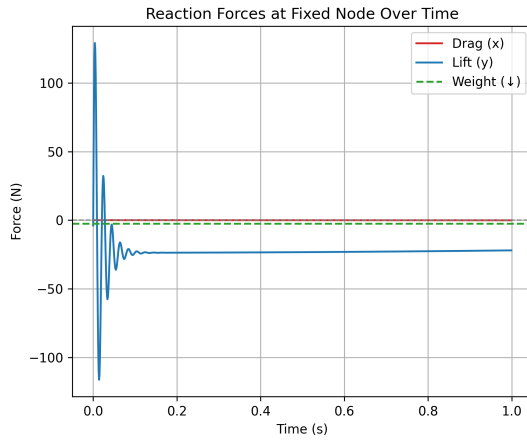


Fig. 4. Case A — Reaction forces at the fixed root during morphing.

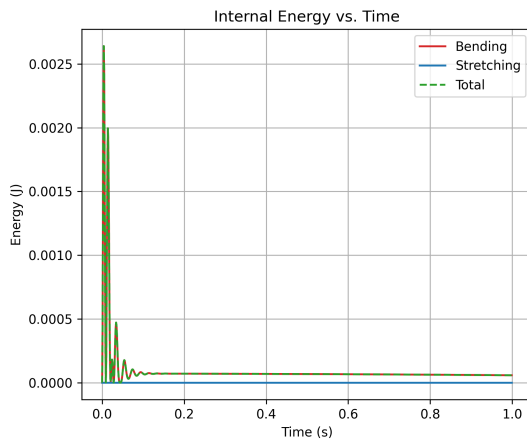


Fig. 5. Case A — Internal bending and stretching energy evolution.

B. Case B — Uniform EI with Aerodynamic Loading

This case retains the same structural setup as Case A (uniform stiffness and exponential curvature) but introduces

aerodynamic loading through a high Reynolds number drag model. A freestream velocity of 20 m/s is applied, generating distributed forces that oppose morphing motion.

This case isolates the effect of aerodynamic drag on structural deformation and energy storage, allowing direct comparison with the baseline structural response.

Observations:

- **Final tip displacement:** $\Delta x = -0.097$ m, $\Delta y = -0.395$ m (total: 0.406 m)
- **Internal energy (final):** Bending: 2.58×10^{-4} J; Stretching: 0 J
- **Peak internal energy:** 2.65×10^{-3} J
- **Reaction forces:** Drag = +0.136 N, Lift = -41.99 N
- **Net lift margin:** -39.35 N

Aerodynamic forces significantly amplify loading compared to Case A, despite a nearly identical final tip displacement (0.406 m). The final bending energy increases by over 4 $\times$ (from 5.9×10^{-5} J to 2.58×10^{-4} J), indicating a much higher steady-state energetic cost.

Reaction forces show the clearest effect of aerodynamic resistance: the drag force reverses direction and increases in magnitude (from -0.095 N to +0.136 N), while lift nearly doubles (from -22.03 N to -41.99 N). When combined with weight, the net lift margin worsens by +20 N, dropping from -19.39 N to -39.35 N.

These results confirm that aerodynamic drag imposes significant energetic and structural penalties, even when deformation is unaffected.

Figures 6–8 visualize the deformation, force response, and energy evolution.

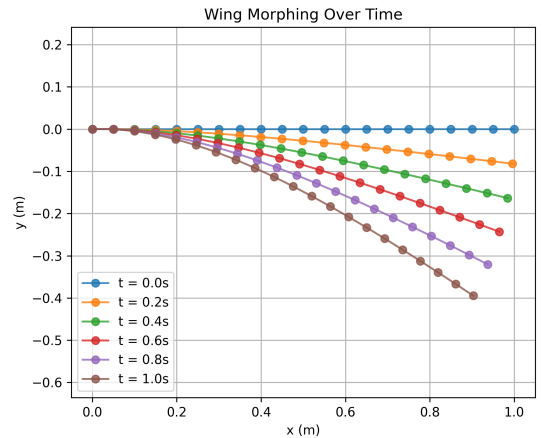


Fig. 6. Case B — Wing morphing over time.

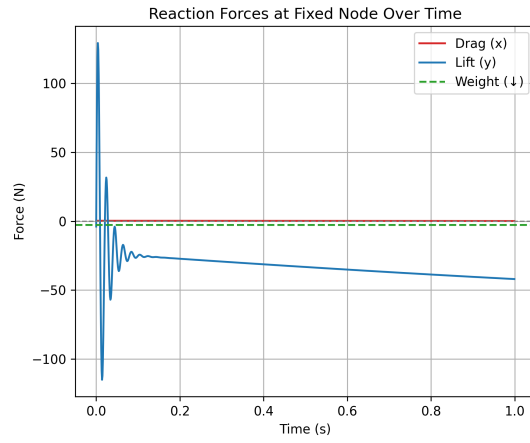


Fig. 7. Case B — Reaction forces at the fixed root.

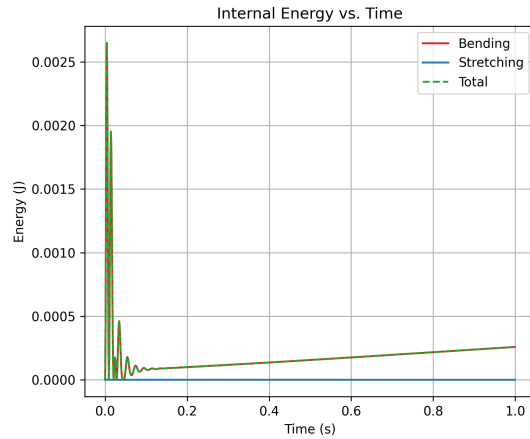


Fig. 8. Case B — Internal energy components over time.

C. Case D — Gaussian Curvature with Uniform Stiffness

This case investigates a biologically inspired curvature profile: a Gaussian distribution centered along the wing chord. The beam retains uniform bending stiffness, and aerodynamic drag is included using a high Reynolds number model. This setup isolates the effect of a more localized curvature actuation compared to the exponential profile in Case B.

Observations:

- **Final tip displacement:** $\Delta x = -0.149$ m, $\Delta y = -0.443$ m (total: 0.467 m)
- **Internal energy (final):** Bending: 2.76×10^{-4} J; Stretching: 0 J
- **Peak internal energy:** 3.26×10^{-3} J
- **Reaction forces:** Drag = +0.289 N, Lift = -41.19 N
- **Net lift margin:** -38.55 N

The Gaussian profile results in significantly higher tip displacement compared to the exponential profile in Case B (0.467 m vs. 0.406 m), confirming the impact of curvature localization. Despite similar lift values, the drag force increases more than 2× from Case B, suggesting greater frontal exposure

during morphing. The final internal energy is slightly higher, and the peak energy increases by 23%, reflecting a higher actuation effort and structural demand.

Figures 9–12 show the curvature profile, deformation, forces, and energy evolving over time.

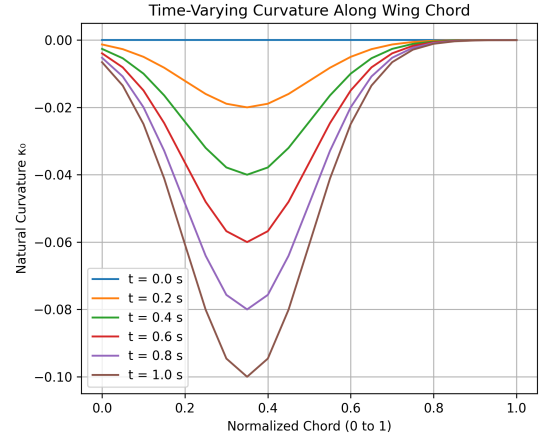


Fig. 9. Case D — Time-varying Gaussian curvature profile concentrated near mid-chord and smoothly ramped to its target value over 1 s.

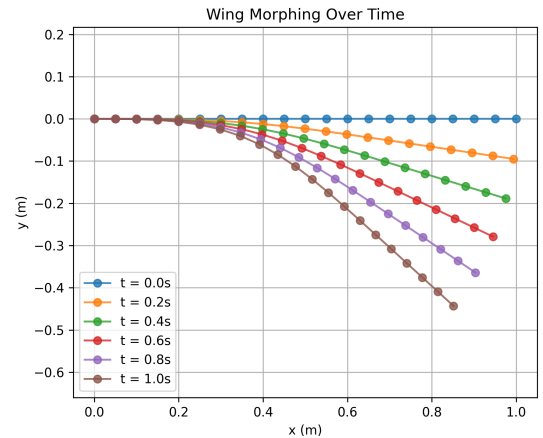


Fig. 10. Case D — Wing morphing under Gaussian curvature.

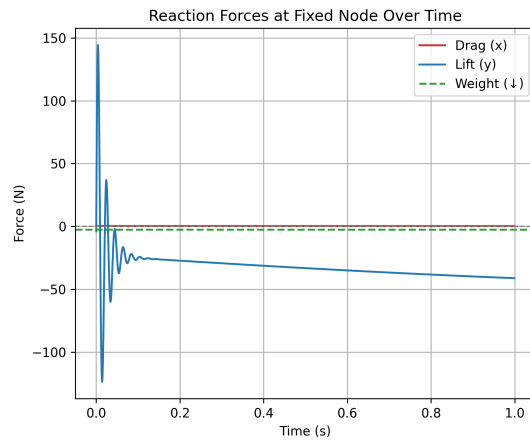


Fig. 11. Case D — Reaction forces at fixed root.

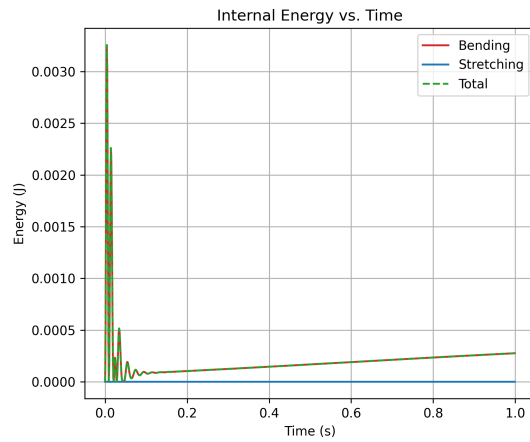


Fig. 12. Case D — Internal energy breakdown over time.

D. Case E — Tip-Softened Wing with Exponential Curvature

In this case, we apply the same exponential curvature as Case B but reduce bending stiffness near the trailing edge (tip-softened stiffness distribution). This simulates a compliant morphing design and allows us to assess how stiffness tapering affects deformation and loading.

Observations:

- **Final tip displacement:** $\Delta x = -0.097$ m, $\Delta y = -0.395$ m (total: 0.406 m)
- **Internal energy (final):** Bending: 2.71×10^{-4} J; Stretching: 0 J
- **Peak internal energy:** 2.65×10^{-3} J
- **Reaction forces:** Drag = +0.136 N, Lift = -41.99 N
- **Net lift margin:** -39.35 N

Despite the tip-softening, the final tip displacement and force response are nearly indistinguishable from Case B. However, the final internal energy is slightly higher, suggesting that passive stiffness variation introduces localized compliance without significantly affecting global response. These results

imply that stiffness tapering can be implemented without compromising structural performance under steady aerodynamic loading.

Figures 13–16 visualize the stiffness profile, deformation, force response, and energy evolution.

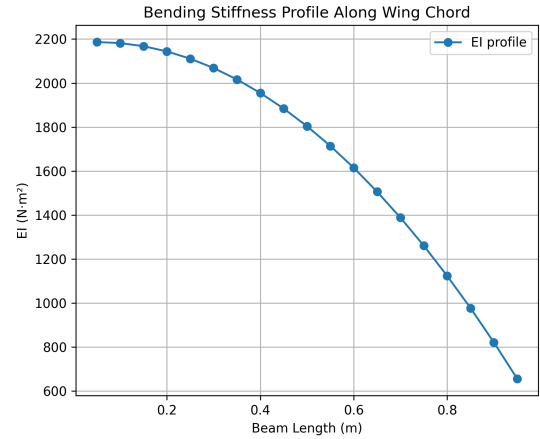


Fig. 13. Case E — Tip-softened bending stiffness (EI) profile with reduced stiffness near the trailing edge.

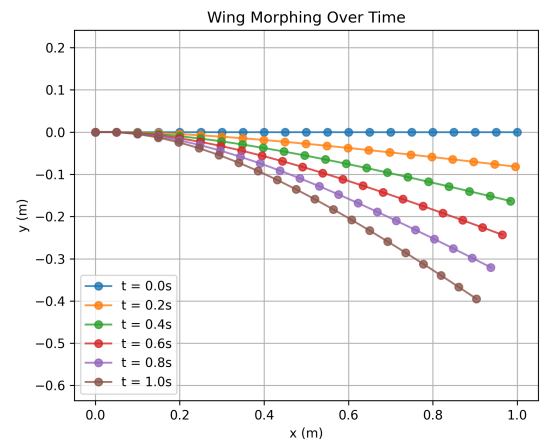


Fig. 14. Case E — Wing morphing with tip-softened stiffness.

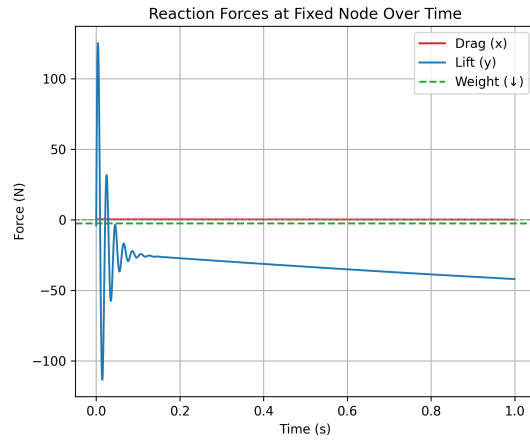


Fig. 15. Case E — Reaction forces at the root.

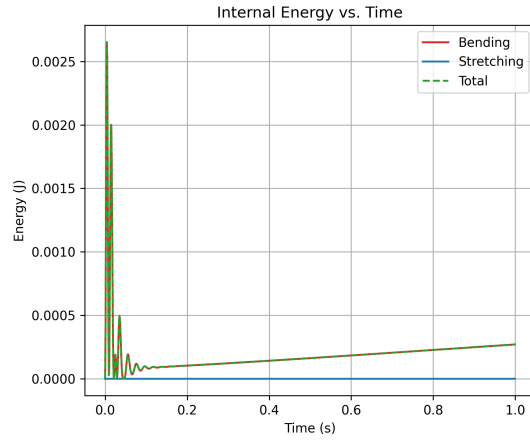


Fig. 16. Case E — Internal energy evolution.

E. Case H — Linearly Tapered Stiffness with Linear Curvature

Case H explores the coupling of two structural variations: a linearly tapered stiffness profile and a linearly increasing curvature field from root to tip. This pairing creates a biologically plausible gradient in both actuation and compliance, enabling large tip deformation.

Observations:

- **Final tip displacement:** $\Delta x = -0.224$ m, $\Delta y = -0.559$ m (total: 0.602 m)
- **Internal energy (final):** Bending: 3.75×10^{-4} J; Stretching: 0 J
- **Peak internal energy:** 6.31×10^{-3} J
- **Reaction forces:** Drag = $+0.092$ N, Lift = -47.20 N
- **Net lift margin:** -44.56 N

This configuration yields the largest deformation of all cases, with over 0.60 m of tip displacement—nearly 50% greater than Case B. However, this enhanced mobility comes at a cost: peak internal energy surges to 6.31×10^{-3} J (2.4× higher than Case B), and lift forces increase by 12%, reaching the highest observed load. This reflects a trade-off between mobility and structural demand.

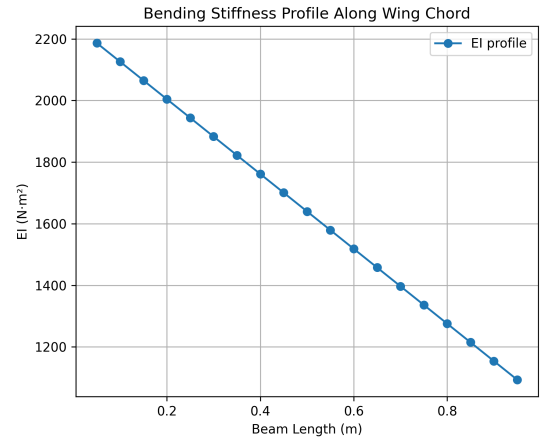


Fig. 17. Case H — Linearly tapered bending stiffness (EI) profile.

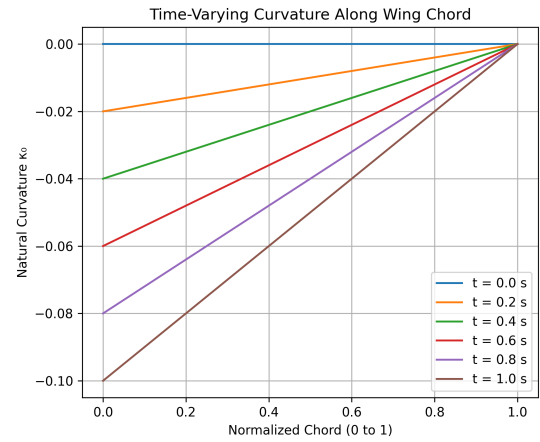


Fig. 18. Case H — Linearly varying prescribed curvature distribution.

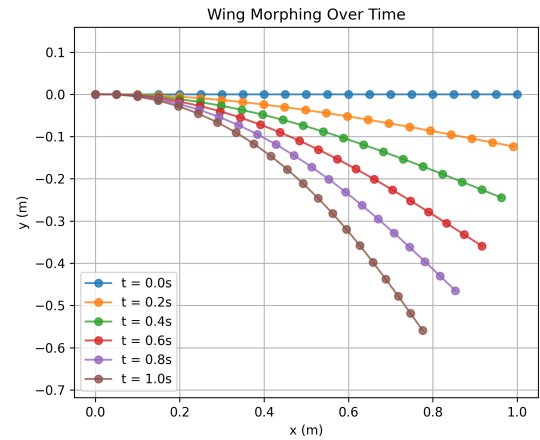


Fig. 19. Case H — Morphing with linearly tapered stiffness.

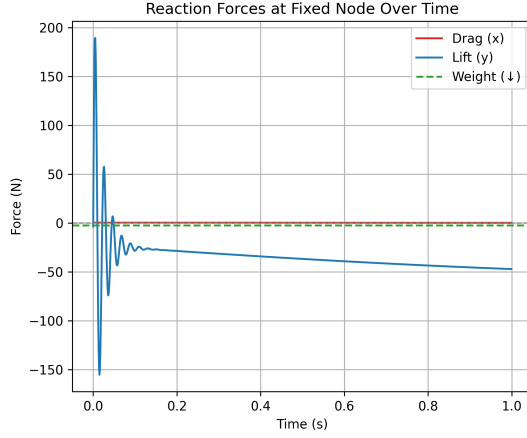


Fig. 20. Case H — Reaction forces.

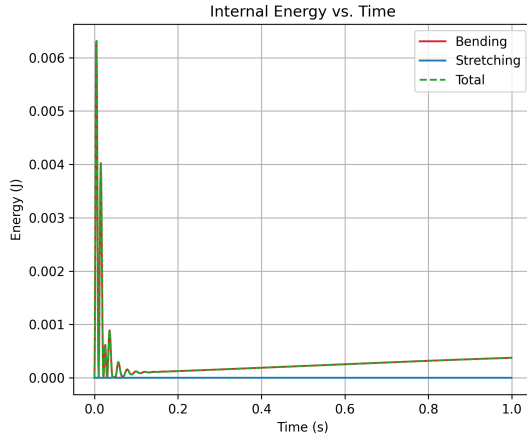


Fig. 21. Case H — Energy usage over time.

VI. DISCUSSION

Across the five simulated morphing configurations, several key insights emerge regarding the interplay between curvature actuation, structural compliance, and aerodynamic loading. Cases B, D, E, and H—all incorporating aerodynamic drag—show that while external airflow has minimal effect on final deformation geometry, it imposes a significant energetic cost and greatly amplifies structural loading at the root.

Comparing Cases A and B (same exponential curvature and stiffness), the final tip displacement remains unchanged at 0.406 m. However, the final bending energy quadruples with aerodynamic forces (from 5.9×10^{-5} J to 2.58×10^{-4} J), and the net lift margin becomes over twice as negative (from -19.39 N to -39.35 N). These results confirm that aerodynamic drag contributes heavily to the steady-state energetic burden despite minimal geometric impact.

The curvature distribution also plays a major role. Case D, using a Gaussian profile, produces a 15% larger tip displacement (0.467 m) than Case B, and the drag force more than doubles ($+0.289$ N vs. $+0.136$ N). This suggests that highly

localized curvature near mid-chord increases frontal area exposure during morphing, exacerbating aerodynamic resistance. The energetic cost also increases modestly, with a 7% rise in final bending energy and a 23% increase in peak energy. This demonstrates a trade-off between enhanced morphing amplitude and actuation cost when using biologically inspired curvature profiles.

Stiffness distribution has a subtler influence. Comparing Cases B and E (same curvature, with tip-softened stiffness in E), tip displacement and aerodynamic forces are virtually identical. However, Case E has a slightly higher final bending energy (2.71×10^{-4} J), suggesting that the added compliance leads to marginally increased energy absorption, likely due to localized flexibility allowing more deformation without appreciable gain in displacement. Thus, stiffness tapering may aid in stress redistribution without altering global response under steady drag.

The most extreme case is H, combining a linear curvature ramp with a linearly tapered stiffness. It results in the highest tip displacement (0.602 m), 50% greater than Case B. However, this comes at the cost of significantly higher structural demand: peak energy is 2.4 $\times$ higher, and the root lift reaction reaches -47.20 N, the largest of all cases. These results illustrate that pairing increasing curvature with decreasing stiffness enables greater deformation but introduces steep energy and load penalties—highlighting a design trade-off between morphing capability and mechanical burden.

Together, these trends suggest that while curvature shaping and stiffness gradients can enhance morphing performance, their benefits must be weighed against energetic cost and increased structural load. Aerodynamic forces, although simplified in this model, are shown to be a primary driver of internal energy demand and load amplification.

VII. CONCLUSION

This study presented a dynamic simulation framework for curvature-driven camber morphing using a reduced-order elastic beam model with variable stiffness and aerodynamic loading. Five morphing configurations were analyzed to investigate how curvature profiles and stiffness gradients interact with aerodynamic forces to influence deformation magnitude, internal energy, and reaction loads.

Key findings:

- Aerodynamic loading substantially increases internal energy and structural forces (e.g., Case B vs. A), even when morphing geometry remains unchanged.
- Localized curvature profiles (e.g., Gaussian) enhance tip deformation but increase drag and energetic cost (Case D).
- Tip-softened stiffness introduces local compliance with minimal effect on global deformation or force response (Case E).
- Linearly tapered stiffness with increasing curvature enables the largest morphing amplitude, but with the highest structural demand (Case H).

TABLE I
COMPARATIVE SUMMARY OF FINAL SIMULATION METRICS

Case	Total Tip Displacement (m)	Drag Force (N)	Net Lift Margin (N)	Final Bending Energy (J)	Peak Bending Energy (J)
A	0.406	−0.095	−19.39	5.9×10^{-5}	2.64×10^{-3}
B	0.406	+0.136	−39.35	2.58×10^{-4}	2.65×10^{-3}
D	0.467	+0.289	−38.55	2.76×10^{-4}	3.26×10^{-3}
E	0.406	+0.136	−39.35	2.71×10^{-4}	2.65×10^{-3}
H	0.602	+0.092	−44.56	3.75×10^{-4}	6.31×10^{-3}

These results confirm that bio-inspired design elements—such as curvature shaping and stiffness gradients—can be leveraged to achieve adaptive wing morphing, but they must be carefully tuned to balance performance with energetic and structural constraints.

Limitations and Assumptions: This study employed a two-dimensional beam model focused on chord-wise deformation. While sufficient to capture key morphing trends, it excludes spanwise twist, 3D effects, and pressure-based aerodynamic forces. The aerodynamic loading model is simplified, omitting lift-curve slope dynamics and unsteady flow behavior. Actuation is modeled as an ideal curvature prescription, without accounting for actuator mass, energy use, or time delay. Future work could incorporate 3D simulations, nonlinear aeroelastic coupling, and feedback control strategies to more accurately model real-world morphing wing systems.

REFERENCES

- [1] D. e. a. Lentink, “How swifts control glide performance with wing shape despite fixed wings,” *Nature*, vol. 446, no. 7139, pp. 1082–1085, 2007.
- [2] S. Barbarino, O. Bilgen, R. M. Ajaj, M. I. Friswell, and D. J. Inman, “A review of morphing aircraft,” *Journal of Intelligent Material Systems and Structures*, vol. 22, no. 9, pp. 823–877, 2011.
- [3] F. Gandhi and P. Anusonti-Inthra, “Design and testing of a variable span morphing wing,” *Journal of Aircraft*, vol. 45, no. 6, pp. 2066–2074, 2009.
- [4] S. Barbarino, E. Saavedra Flores, R. M. Ajaj, I. Dayyani, and M. I. Friswell, “A review on shape memory alloys with applications to morphing aircraft,” *Smart Materials and Structures*, vol. 23, no. 6, p. 063001, 2014.
- [5] A. Y. N. Sofla, S. A. Meguid, K. T. Tan, and W. K. Yeo, “Shape morphing of aircraft wing: Status and challenges,” *Materials & Design*, vol. 31, no. 3, pp. 1284–1292, 2010.
- [6] C. Thill, J. Etches, I. Bond, K. Potter, and P. Weaver, “Morphing skins,” *The Aeronautical Journal*, vol. 112, no. 1129, pp. 117–139, 2008.
- [7] S. Joshi, Z. Tidwell, and P. Flick, “Aeroelastic modeling of a morphing wing using finite elements,” in *45th AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics and Materials Conference*, 2004.
- [8] K. Mohseni and R. Mittal, “Feedback control of vortex shedding using fluidic actuation,” *AIAA Journal*, vol. 47, no. 5, pp. 1160–1170, 2009.
- [9] J. Chen and H. A. Bruck, “Design of morphing airfoils using compliant mechanisms,” *Journal of Intelligent Material Systems and Structures*, vol. 21, no. 12, pp. 1189–1201, 2010.
- [10] U. Icardi and L. Ferrero, “Preliminary study of an adaptive wing with shape memory alloy torsion actuators,” *Materials & Design*, vol. 30, no. 10, pp. 4200–4210, 2009.

APPENDIX

TABLE II
SUMMARY OF PHYSICAL AND SIMULATION PARAMETERS

Parameter	Value	Justification
Beam length L	1.0 m	Normalized unit chord length to simplify parameter sweeps.
Number of nodes n_v	21	Balances resolution with computational efficiency.
Element spacing ΔL	0.05 m	Uniform discretization along chord.
Young's modulus Y	200 GPa	Representative of carbon fiber composites.
Density ρ	1700 kg/m ³	Matches carbon-fiber-like density for aerospace materials.
Inner radius r_0	11 mm	Hollow beam cross-section.
Outer radius R	13 mm	Outer boundary of beam cross-section.
Cross-sectional area A	$\pi(R^2 - r_0^2)$	For computing mass and axial stiffness.
Second moment of area I	$\frac{\pi}{4}(R^4 - r_0^4)$	For bending stiffness calculations.
Gravitational acceleration g	9.81 m/s ²	Used to apply beam weight.
Air density ρ_{air}	1.2 kg/m ³	Standard at sea level.
Dynamic viscosity μ	1.8×10^{-5} Pa s	Approximate value for air at 20°C.
Drag coefficient C_d	1.2	Typical for bluff bodies under high-Re conditions.
Tangential/Normal resistance (C_t, C_p)	(2.0, 10.0)	For low-Re drag models; unused in this study.
Airflow velocity $v_{forward}$	0 or 20 m/s	Examines cases with and without aerodynamic drag.
Drag model	none / highRe	Allows study of structural-only and aeroelastic behavior.
Curvature profile $\kappa_0(s)$	exponential / linear / gaussian / uniform	Shapes selected to reflect engineering and biological patterns.
Max curvature magnitude κ_{max}	-0.1	Chosen to ensure physically realistic actuation.
Time step Δt	0.001 s	Sufficient for capturing dynamic effects with stability.
Simulation duration T	1.0 s	Matches curvature actuation ramp.